Sports Day Medal Combinations - Claude

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A student participates in **four** different sports on Sport’s Day at her school. Each sport has three medals assigned, **gold**, **silver**, and **bronze**. It is possible that the student wins no medals. How many possible ways can she get **at least one** medal?

I need to find the number of ways a student can get at least one medal across four sports, where each sport has three possible medals (gold, silver, bronze).

Let me think about this step by step.

First, I'll identify what's possible for each sport:

- Win gold medal
- Win silver medal
- Win bronze medal
- Win no medal

So for each sport, there are 4 possible outcomes.

Since there are 4 sports total, the total number of possible outcomes is: 4^4 = 256

Now I need to find the number of ways to get "at least one medal." It's easier to use the complement: find the number of ways to get NO medals at all, then subtract from the total.

The number of ways to get no medals:

- For each sport, there's exactly 1 way to get no medal
- Across all 4 sports: 1^4 = 1 way to get no medals at all

Therefore, the number of ways to get at least one medal is: Total outcomes - Ways to get no medals = 256 - 1 = **255**

The answer is 255 possible ways.

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why 4^4

Good question! Let me break down why it's 4^4.

For each individual sport, the student has these possible outcomes:

1. Win gold medal
2. Win silver medal
3. Win bronze medal

4. Win no medal

So each sport has **4 possible outcomes**.

Since the student participates in **4 different sports**, and the outcome in each sport is independent of the others, I use the multiplication principle:

- Sport 1: 4 possible outcomes
- Sport 2: 4 possible outcomes
- Sport 3: 4 possible outcomes
- Sport 4: 4 possible outcomes

Total combinations = 4 × 4 × 4 × 4 = 4^4 = 256

This counts every possible combination, like:

- (gold, silver, no medal, bronze)
- (no medal, no medal, gold, no medal)
- (bronze, bronze, bronze, bronze)
- (no medal, no medal, no medal, no medal)
- etc.

The exponent 4 comes from having 4 sports, and the base 4 comes from having 4 possible outcomes per sport.

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4 × 4 × 4 × 4 we multiply why

We multiply because of the **multiplication principle** (also called the fundamental counting principle).

Here's why we multiply instead of add:

Each choice is independent and sequential

Think of it this way - for every single outcome in Sport 1, the student can have ANY of the 4 outcomes in Sport 2.

Let me illustrate with a smaller example. Say there are only 2 sports, each with 2 possible outcomes (medal or no medal):

Sport 1 → Sport 2:

- Medal → Medal
- Medal → No medal
- No medal → Medal
- No medal → No medal

That's 2 × 2 = 4 total combinations.

For our actual problem:

- For each of the 4 outcomes in Sport 1, there are 4 possible outcomes in Sport 2
- For each of those 4 × 4 = 16 combinations of Sports 1&2, there are 4 possible outcomes in Sport 3
- For each of those 4 × 4 × 4 = 64 combinations of Sports 1&2&3, there are 4 possible outcomes in Sport 4

This gives us 4 × 4 × 4 × 4 = 256 total combinations.

Why not addition? If we added (4 + 4 + 4 + 4 = 16), we'd only be counting the individual possibilities for each sport separately, not all the ways they can be combined together.

The multiplication captures all possible **combinations** across all sports.